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# 3.1 Ratio and Proportion

OCR GCSE Mathematics J560 | Component 3 (Ratio, Proportion and Rates of Change) | Papers 1, 2 & 3

## 3.1.1 Simplifying and Using Ratios

**Ratio** — a ratio compares two or more quantities of the *same type* and shows their relative sizes. It is written using a colon, e.g. $3:5$.

### Writing a Ratio

- Quantities must be in the **same units** before forming a ratio.
- Order matters: "the ratio of boys to girls is $2:3$" is different from $3:2$.
- Ratios have **no units** — they are pure numbers once units are matched.

**Worked Example:** Write the ratio 40 cm to 1.2 m in its simplest form.

**Step 1:** Convert to the same unit: $1.2\,\mathrm{m} = 120\,\mathrm{cm}$
**Step 2:** Write the ratio: $40:120$
**Step 3:** Simplify by dividing both by HCF (40): $1:3$

### Simplifying to Simplest Form

- Divide all parts of the ratio by their **Highest Common Factor (HCF)**.
- A ratio is in simplest form when all parts are integers with HCF = 1.
- For ratios with fractions or decimals, multiply through to clear them first.

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**Worked Example:** Simplify the ratio $1.5 : 4.5$.

Multiply both by 2 to clear decimals: $3 : 9$
Divide by HCF (3): $\mathbf{1 : 3}$

### Equivalent Ratios

Equivalent ratios are found by multiplying or dividing all parts by the same non-zero number, just like equivalent fractions.

| Ratio | Multiply by 2 | Multiply by 5 |
|---|---|---|
| $1 : 3$ | $2 : 6$ | $5 : 15$ |
| $2 : 5$ | $4 : 10$ | $10 : 25$ |

**Exam Tip:** Always check units match before writing a ratio. A common mistake is to write "30 minutes to 2 hours" as $30 : 2$ instead of $30 : 120 = 1 : 4$.

## 3.1.2 Dividing in a Ratio

**Dividing in a ratio** — split a total quantity into parts proportional to a given ratio.

### Method

1. Find the **total number of parts** (add the ratio values).
2. Find the **value of one part** by dividing the total by the number of parts.
3. Multiply by each ratio value to find each share.

**Worked Example:** Share £420 in the ratio $3 : 4$.

**Step 1:** Total parts = $3 + 4 = 7$
**Step 2:** Value of 1 part = $£420 \div 7 = £60$
**Step 3:** First share = $3 \times £60 = £180$; Second share = $4 \times £60 = £240$
**Check:** $£180 + £240 = £420$ ✓

**Worked Example:** A recipe uses flour, butter and sugar in the ratio $5 : 3 : 2$. If the total mixture is 600 g, find the mass of each ingredient.

Total parts = $5 + 3 + 2 = 10$
1 part = $600 \div 10 = 60\,\text{g}$
Flour: $5 \times 60 = 300\,\text{g}$
Butter: $3 \times 60 = 180\,\text{g}$
Sugar: $2 \times 60 = 120\,\text{g}$

**Worked Example:** Alice and Bob share profits in the ratio $2 : 5$. Alice receives £140. How much does Bob receive?

Alice's 2 parts = £140, so 1 part = £70
Bob's share = $5 \times £70 = \mathbf{£350}$

**Exam Tip:** When only one share is given, find "1 part" first, then scale to the other share. You do not need to find the total first.

## 3.1.3 Comparing Using Ratios

### Unit Ratio (1 : n or n : 1)

A unit ratio has 1 as one of its parts. It makes comparison easy.

**Worked Example:** Write $6 : 15$ as a unit ratio of the form $1 : n$.

Divide both by 6: $1 : 2.5$

### Best Buy Problems

To find the best value, compare the **price per unit** (unit rate). Divide the price by the quantity for each option.

**Worked Example:** Box A: 400 g for £1.60. Box B: 550 g for £2.09. Which is the better value?

Box A: $£1.60 \div 400 = £0.0040$ per gram
Box B: $£2.09 \div 550 = £0.0038$ per gram
Box B is cheaper per gram — **Box B is better value**.

**Exam Tip:** Always state your conclusion clearly. Show the unit rate for each option so the examiner can award method marks even if arithmetic slips.

## 3.1.4 Direct Proportion

Two quantities are in **direct proportion** if, when one doubles, the other doubles (and so on). Their ratio remains constant. We write $y \propto x$, which means $y = kx$ for some constant $k$.

### Finding the Constant of Proportionality

1. Use a known pair of values to find $k = y/x$.
2. Write the equation $y = kx$.
3. Use the equation to find unknown values.

**Worked Example:** $y$ is directly proportional to $x$. When $x = 4$, $y = 20$. Find $y$ when $x = 7$.

$k = y/x = 20/4 = 5$

Equation: $y = 5x$
When $x = 7$: $y = 5 \times 7 = \mathbf{35}$

**Worked Example:** 5 identical books cost £12.50. Find the cost of 8 books.

Cost per book = $£12.50 \div 5 = £2.50$
Cost of 8 books = $8 \times £2.50 = \mathbf{£20.00}$

### Graph of Direct Proportion

A direct proportion graph is a **straight line through the origin**. The gradient equals $k$.

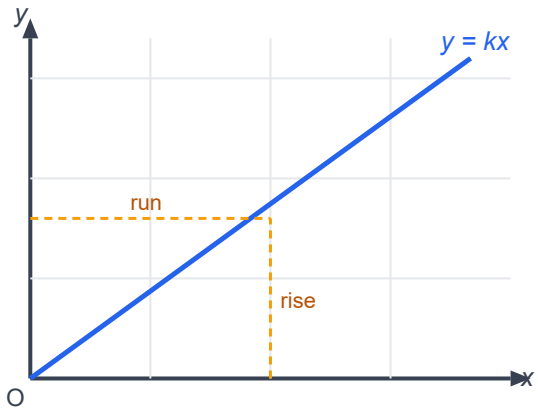


Fig 1 — Direct proportion: straight line through origin with gradient $k$

## 3.1.5 Inverse Proportion

Two quantities are in **inverse proportion** if, when one doubles, the other halves. We write $y \propto \frac{1}{x}$, which means $y = \frac{k}{x}$ for some constant $k$. The product $xy$ is always constant.

### Finding k for Inverse Proportion

1. Use a known pair of values: $k = xy$.
2. Write the equation $y = k/x$.
3. Substitute to find unknowns.

**Worked Example:** $y$ is inversely proportional to $x$. When $x = 3$, $y = 8$. Find $y$ when $x = 6$.

$k = xy = 3 \times 8 = 24$
Equation: $y = 24/x$
When $x = 6$: $y = 24/6 = \mathbf{4}$

**Worked Example:** 8 workers take 15 days to complete a project. How long would 12 workers take (assuming constant rate)?

$k = 8 \times 15 = 120$ (worker-days)
Time = $120 \div 12 = \mathbf{10}$ days

## Higher: Other Proportionality Relationships Higher

| Type | Notation | Equation |
|---|---|---|
| Direct (square) | $y \propto x^2$ | $y = kx^2$ |
| Direct (cube) | $y \propto x^3$ | $y = kx^3$ |
| Inverse (square) | $y \propto \frac{1}{x^2}$ | $y = \frac{k}{x^2}$ |
| Inverse (square root) | $y \propto \frac{1}{\sqrt{x}}$ | $y = \frac{k}{\sqrt{x}}$ |

**Worked Example:** $y \propto x^2$. When $x = 3$, $y = 36$. Find $y$ when $x = 5$.

$k = y/x^2 = 36/9 = 4$
Equation: $y = 4x^2$
When $x = 5$: $y = 4 \times 25 = \mathbf{100}$

**Worked Example:** $y \propto \frac{1}{x^2}$. When $x = 2$, $y = 5$. Find $y$ when $x = 10$.

$k = yx^2 = 5 \times 4 = 20$
Equation: $y = 20/x^2$
When $x = 10$: $y = 20/100 = \mathbf{0.2}$

### Proportion Graphs: Direct vs Inverse

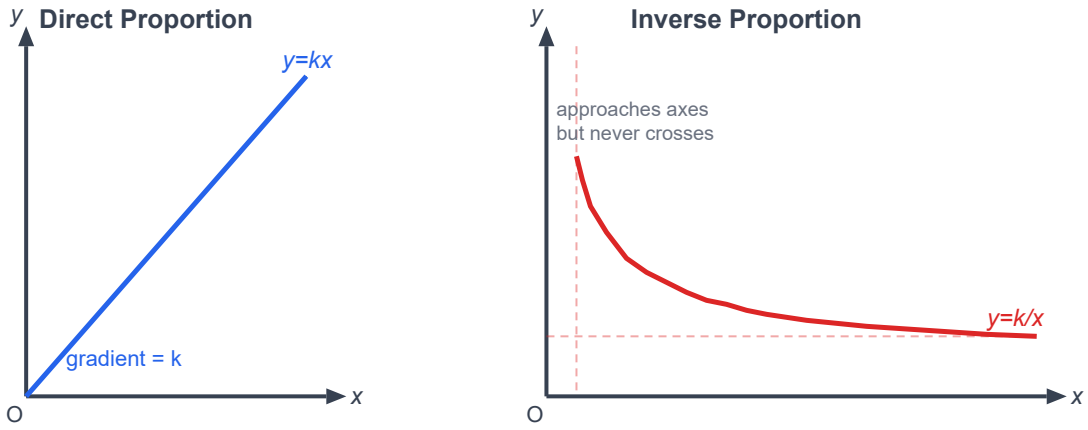


Fig 2 — Left: direct proportion ($y = kx$, straight line through origin). Right: inverse proportion ($y = k/x$, hyperbola approaching both axes).

**Exam Tip:** The inverse proportion graph never touches the axes. If asked to sketch it, draw a curve that gets closer and closer to each axis (asymptotic behaviour).

## 3.1.6 Proportionality in Geometry — Similar Shapes

**Similar shapes** have the same angles and all corresponding lengths are in the same ratio (the *scale factor* $k$). They are enlargements of each other.

### Linear Scale Factor

If shape B is an enlargement of shape A with linear scale factor $k$, then every length in B is $k$ times the corresponding length in A.

$$k = \frac{\text{length in image}}{\text{corresponding length in original}}$$

## Area Scale Factor and Volume Scale Factor

If the linear scale factor is $k$, then:

$$\text{Area scale factor} = k^2$$

$$\text{Volume scale factor} = k^3$$

| Relationship | Scale Factor |
|---|---|
| Lengths (1D) | $k$ |
| Areas (2D) | $k^2$ |
| Volumes (3D) | $k^3$ |

**Worked Example:** Two similar triangles have corresponding sides of 4 cm and 10 cm. (a) Find the area scale factor. (b) If the smaller triangle has area 24 cm$^2$, find the area of the larger.

Linear scale factor: $k = 10/4 = 2.5$
(a) Area scale factor: $k^2 = 2.5^2 = 6.25$
(b) Area of larger = $24 \times 6.25 = \mathbf{150}\,\mathrm{cm}^{\mathbf{2}}$

**Worked Example:** Two similar cylinders have heights 6 cm and 9 cm. The smaller has volume 80 cm$^3$. Find the volume of the larger.

Linear scale factor: $k = 9/6 = 1.5$
Volume scale factor: $k^3 = 1.5^3 = 3.375$
Volume of larger = $80 \times 3.375 = \mathbf{270}\,\mathrm{cm}^{\mathbf{3}}$

**Worked Example:** Two similar shapes have areas 50 cm² and 200 cm². Find the ratio of their perimeters.

Area scale factor: $200/50 = 4$
Linear scale factor: $k = \sqrt{4} = 2$
Ratio of perimeters = $1 : 2$

**Exam Tip:** Always identify what you are given — length, area or volume — and apply the correct power. If you are given areas and want lengths, take the square root. If you are given volumes and want lengths, take the cube root.

## Formula Summary

| Topic | Key Formula / Rule |
|---|---|
| Simplify ratio | Divide all parts by HCF; match units first |
| Divide in ratio $a : b$ | 1 part = Total $\div(a + b)$; shares = $a\times$part and $b\times$part |
| Direct proportion | $y = kx$; $k = y/x$ is constant |
| Inverse proportion | $y = k/x$; $k = xy$ is constant |
| $y \propto x^2$ (Higher) | $y = kx^2$; $k = y/x^2$ |
| $y \propto 1/x^2$ (Higher) | $y = k/x^2$; $k = yx^2$ |
| Similar shapes — area | Area SF $= k^2$ |
| Similar shapes — volume | Volume SF $= k^3$ |

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